Gradient-intercept form

1

a Gradient

b y-intercept

2

a The constant term is the same.

b All the graphs intersect the y-axis at the same point.

c The lines will pass through $(0, 1)$.

3

a The coefficient of x is the same.

b All the graphs have the same gradient.

c The line will have gradient 2.

4

a Equal

b Not equal

c Not equal

d Equal

5

a $y = 3$

b $m = 2$

6

a -4

b $m = -4$

7

a No

b No

c No

d Yes

e Yes

f No

g Yes

h Yes

8

a

i $m = -2$ **ii** $c = 9$

b

i $m = -4$ **ii** $c = -8$

c

i $m = 8$ **ii** $c = 6$

d

i $m = \frac{9}{2}$ **ii** $c = -1$

e

i $m = 4$ **ii** $c = -5$

f

i $m = -1$ **ii** $c = 3$

g

i $m = \frac{3}{10}$ **ii** $c = \frac{1}{5}$

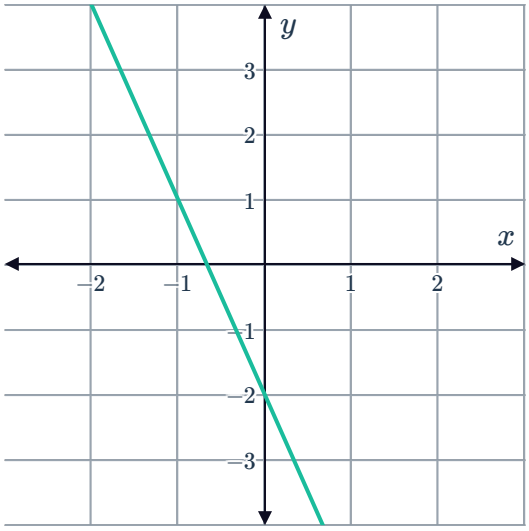
h

i $m = -\frac{4}{3}$ **ii** $c = \frac{5}{6}$

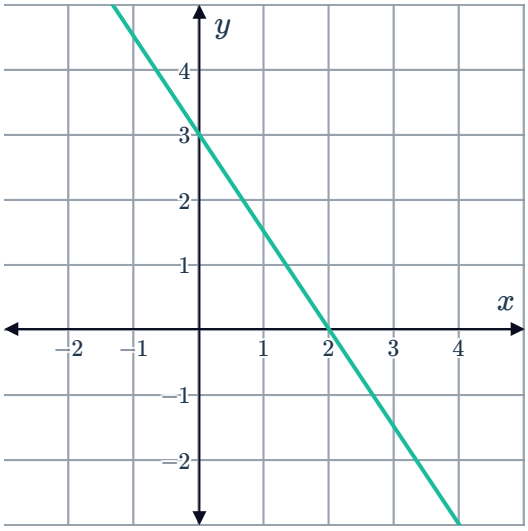
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9

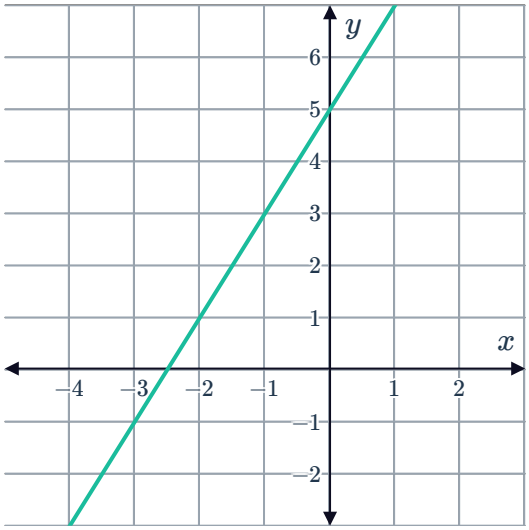
a



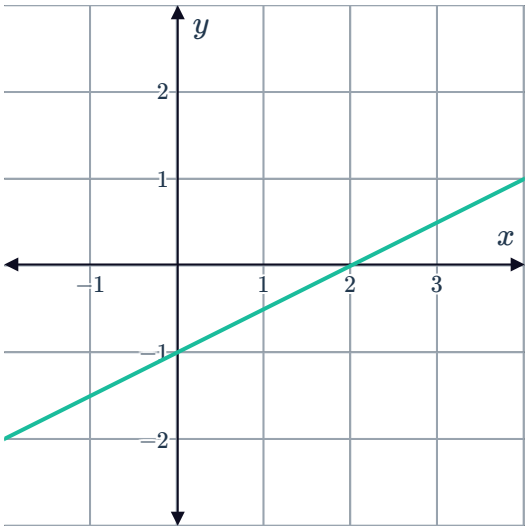
b



c



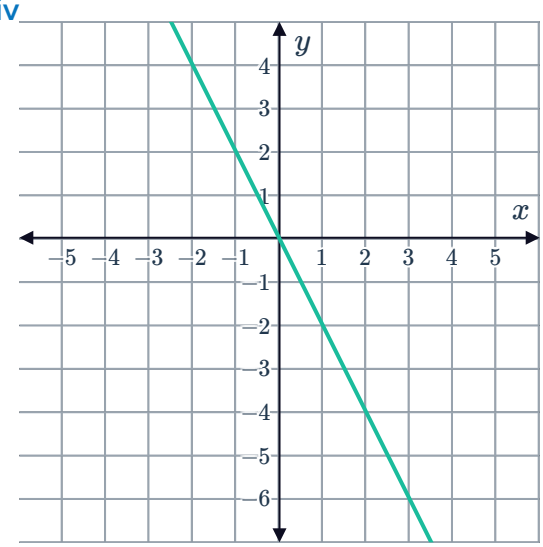
d



10

a

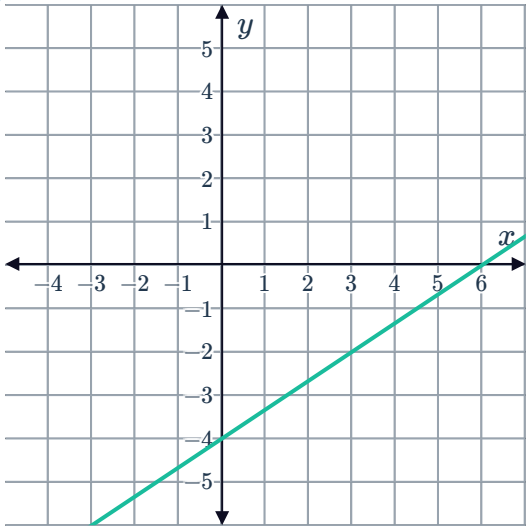
i $y = 0$ **ii** $x = 0$ **iii** $y = -6$ **iv**



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

b

i $y = -4$ ii $x = 6$ iii $y = -2$ iv



11

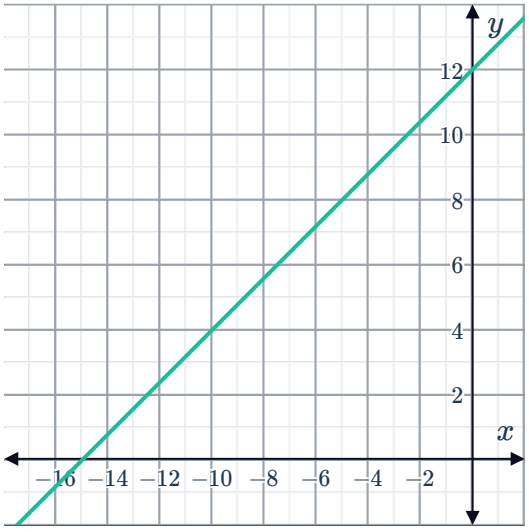
a

i $y = 12$

ii $y = \frac{4x}{5} + 12$

iii $x = -15$

iv



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

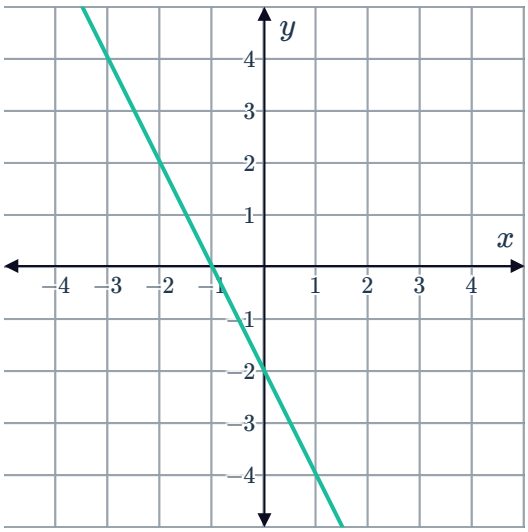
b

i $y = -2$

ii $y = -2x - 2$

iii $x = -1$

iv



Equation of a line

12

a $y = -3x - 2$

b Yes

13

a $y = -3x + 8$

b No

14

a $y = -8x - 3$

b $y = 2x + 5$

c $y = -6x + 9$

d $y = -8x - 9$

e $y = -\frac{3x}{4} + 3$

f $y = \frac{4x}{3} + 3$

g $y = -2x - 15$

h $y = 8x - 4$

i $y = \frac{2}{13}$

15

a

i $m = -3$

ii $c = 4$

b $y = -3x + 4$

c $y = -77$

16

a $y = 3x - 4$

b $y = 146$

17

a $y = 3x - 4$

b $y = 59$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18

a $y = -3x + 4$ **b** $y = \frac{x}{5} - 3$ **c** $y = \frac{3x}{5} - 2$ **d** $y = \frac{2x}{5} + 2$

e $y = 3x + 1$ **f** $y = -2x + 5$ **g** $y = \frac{x}{2} - 1$ **h** $y = -\frac{5x}{2} - 3$

19

a
- i 2
- ii $y = 2x + 2$

b
- i 2
- ii $y = 2x - 9$

c
- i -6
- ii $y = -6x + 2$

d
- i 7
- ii $y = 7x - 3$

20

a
- i $y = -x + 4$
- ii $m = -1$
- iii $c = 4$

b
- i $y = 9x - 8$
- ii $m = 9$
- iii $c = -8$

c
- i $y = 12x - 9$
- ii $m = 12$
- iii $c = -9$

d
- i $y = 18x - 12$
- ii $m = 18$
- iii $c = -12$

e
- i $y = \frac{x}{3} - 3$
- ii $m = \frac{1}{3}$
- iii $c = -3$

f
- i $y = \frac{3x}{4} - 7$
- ii $m = \frac{3}{4}$
- iii $c = -7$

21

a $y = -\frac{5}{3}x - \frac{5}{3}$ and $y = -\frac{7}{6} + \frac{1}{2}$ **b** Line A

22 $4x + 4y + 1 = 0$

23

a $y = -x + 2$ **b** $a = 8$

Gradients of parallel lines

24

a True **b** False **c** False **d** True

e True

25

a Yes **b** No **c** Yes **d** Yes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

26 **a** -2 **b** 4

27 **a** Yes **b** No **c** Yes **d** No

28 **a** No **b** No **c** No **d** Yes

Point-gradient formula

29 **a** i 3 ii 4 **b** i 0 ii 0 **c** i $\frac{5}{2}$ ii -4 **d** i 5 ii 3

e i 0 ii 0 **f** i $\frac{3}{2}$ ii -3

30 **a**

i $m = 4$

ii $c = 0$

iii $y = 4x$

iv

x	0	1	2	3	4	24
y	0	4	8	12	16	96

b

i $m = 5$

ii $c = 9$

iii $y = 5x + 9$

iv

x	0	1	2	3	4	21
y	9	14	19	24	29	114

c

i $m = 2$

ii $c = -23$

iii $y = 2x - 23$

iv

x	0	1	2	3	4	25
y	-23	-21	-19	-17	-15	27

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

d

i $m = -5$

ii $c = 27$

iii $y = -5x + 27$

iv

x	0	1	2	3	4	70
y	27	22	17	12	7	-323

31

a

i $y = 5x$

ii

x	1	2	3	4	19
y	5	10	15	20	95

b

i $y = -3x$

ii

x	1	2	3	4	16
y	-3	-6	-9	-6	-48

c

i $y = 3x - 2$

ii

x	1	2	3	4	60
y	1	4	7	10	178

d

i $y = -8x - 6$

ii

x	1	2	3	4	80
y	-14	-22	-30	-38	-646

e

i $y = -8x + 90$

ii

x	1	2	3	4	-15
y	82	74	66	58	210

32

a $y = -3x + 4$ **b** $y = 3x + 2$ **c** $y = 5x + 3$ **d** $y = -2x + 5$

33

a $y = -4x - 24$ **b** $y = 2x - \frac{12}{5}$ **c** $y = 4x + 19$ **d** $y = -3x + 15$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

e $y = \frac{25}{2} - \frac{5x}{2}$ **f** $y = -\frac{10x}{3} + \frac{49}{3}$ **g** $y = -9x - 33$ **h** $y = 7x + \frac{98}{9}$

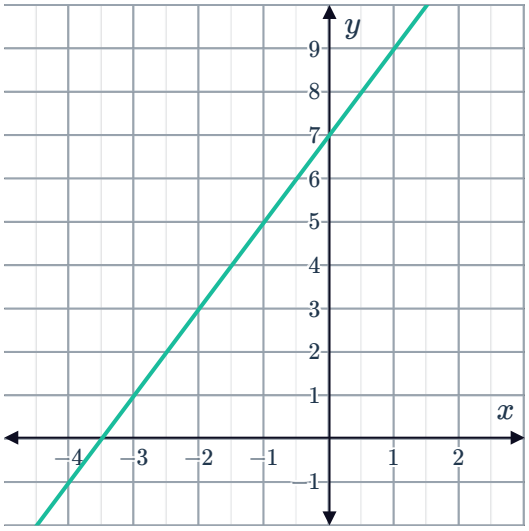
i $y = -19 + \frac{5x}{2}$ **j** $y = \frac{7x}{2} + 19$

34

a

i $y = 2x + 7$

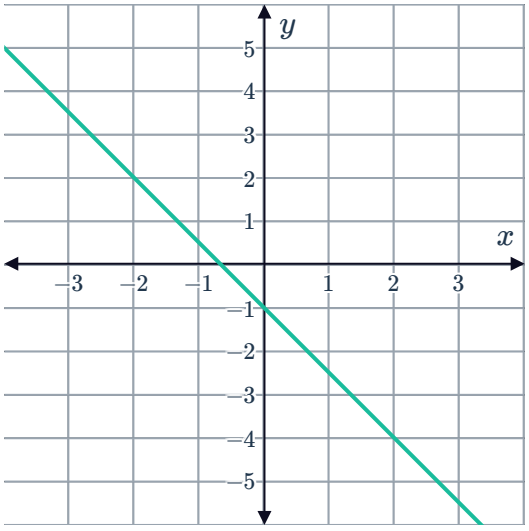
ii



b

i $y = -\frac{3x}{2} - 1$

ii

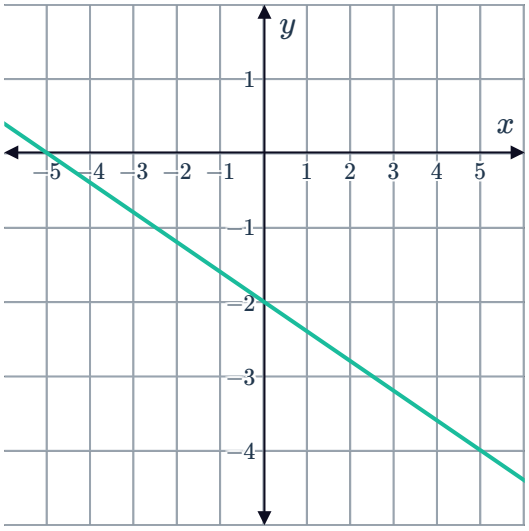


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c

i $y = -\frac{2x}{5} - 2$

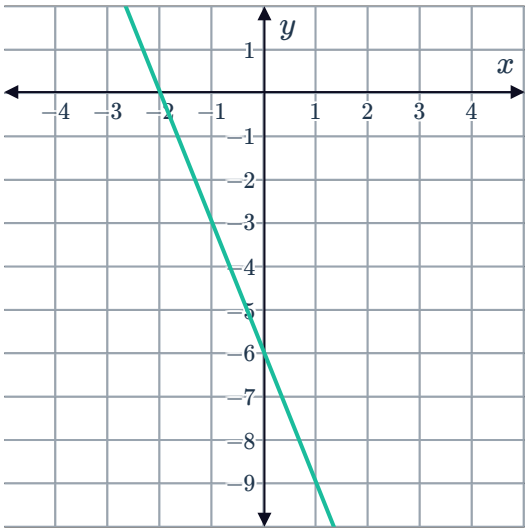
ii



d

i $y = -3x - 6$

ii



35

a $x = 4$

b $y = -4x + 16$

36 $4x - 7y + 40 = 0$

37

a

i $-\frac{13}{7}$

ii $y = -\frac{13x}{7} - \frac{23}{7}$

b

i -4

ii $y = -4x + 9$

c

i $\frac{11}{2}$

ii $11x - 2y + 80 = 0$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

38 **a** Yes **b** No **c** Yes **d** No

39 Example answers:

- $y = 2x + 1$
- $y = 3x$
- $y = -x + 4$

All lines have different gradients and $y-$intercepts.

Applications

40

a $(1, 1)$

b $3x - 10y + 7 = 0$

41

a $(0, 2)$

b $x - 3y + 6 = 0$

42

a $p = 17$

b $q = 12$

c $9x - 2y - 129 = 0$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647